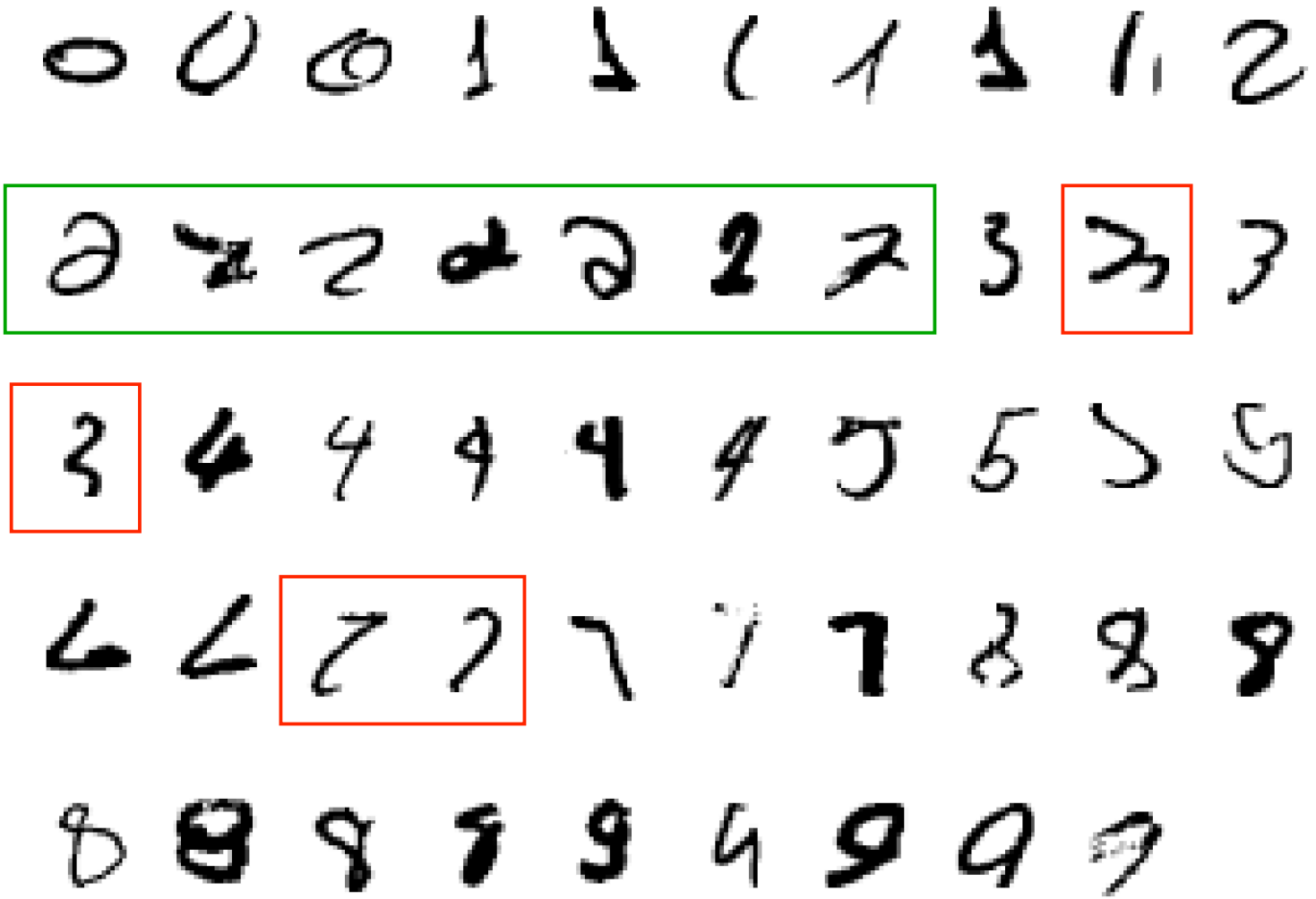
What is Machine Learning?

- How can we solve a specific problem?
  - As computer scientists we write a program that encodes a set of rules that are useful to solve the problem



Figure : How can we make a robot cook?

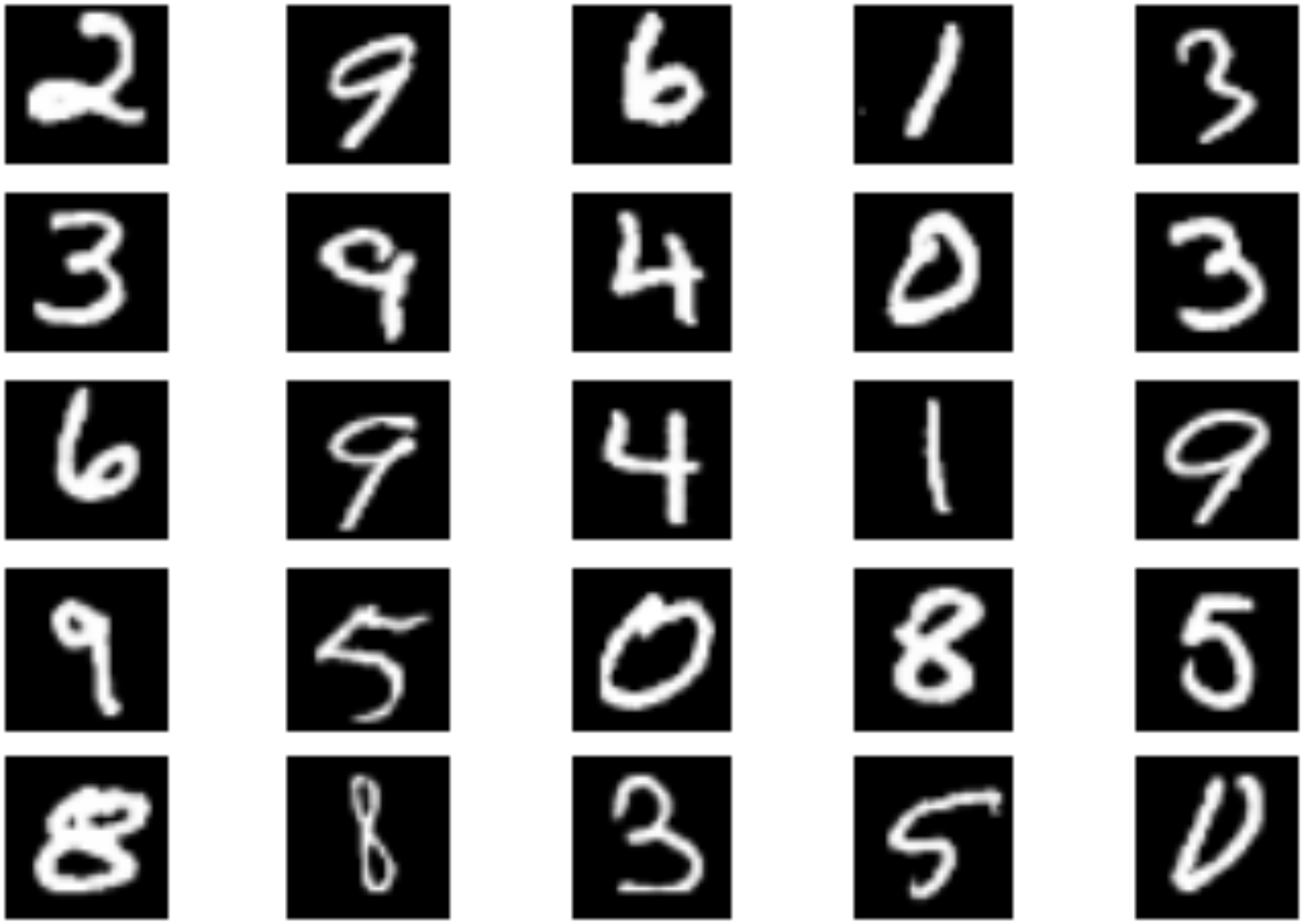
Tasks that requires machine learning: What makes a 2?



Learning algorithms are useful in many tasks

1. Classification: Determine which discrete category the example is

Examples of Classification





Learning algorithms are useful in many tasks

1. Classification: Determine which discrete category the example is
2. Recognizing patterns: Speech Recognition, facial identity, etc

Examples of Recognizing patterns



Figure : Siri: https://www.youtube.com/watch?v=8ciagGASro0

Learning algorithms are useful in other tasks

1. Classification: Determine which discrete category the example is
2. Recognizing patterns: Speech Recognition, facial identity, etc
3. Recommender Systems: Noisy data, commercial pay-off (e.g., Amazon, Netflix).

Examples of Recommendation systems



Learning algorithms are useful in other tasks

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3. Recommender Systems: Noisy data, commercial pay-off (e.g., Amazon, Netflix).
4. Information retrieval: Find documents or images with similar content

Examples of Information Retrieval

Google csc411

Web Maps Videos News Images More Search tools

About 24,500 results (0.52 seconds)

[PDF] CSC 411 MACHINE LEARNING and DATA MINING ...
www.cs.toronto.edu/~zemel/documents/411/syl.pdf
CSC 411. MACHINE LEARNING and DATA MINING. Lectures: Monday, Wednesday 12-1 (section 1), 3-4 (section 2). Lecture Room: MP 134 (section 1); Bahen ...

Professor Richard Zemel - Department of Computer Science
www.cs.toronto.edu/~zemel/
Image Question Answering: A Visual Semantic Embedding Model and a New Dataset . Mengye Ren, Ryan Kiros, Richard Zemel. ICML 2015 Deep Learning ...
Course Offerings - Research Interests - Students & Post Docs - Contact Info

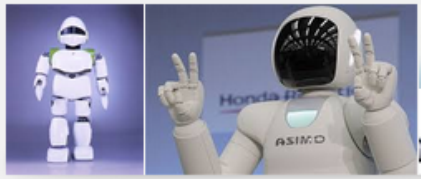
UofT Machine Learning | Course
learning.cs.toronto.edu/courses
CSC 411, Machine Learning and Data Mining (Raquel Urtasun and Richard Zemel); STA 4513, Statistical models of networks, graphs, and other relational ...

CSC 411: Machine Learning and Data Mining
www.cs.utoronto.ca/~radford/**csc411**.F06/
CSC 411: Machine Learning and Data Mining (Sept-Dec 2006). Note: The test on December 8 at 3pm will be held in BA B024, not the usual lecture/tutorial room.

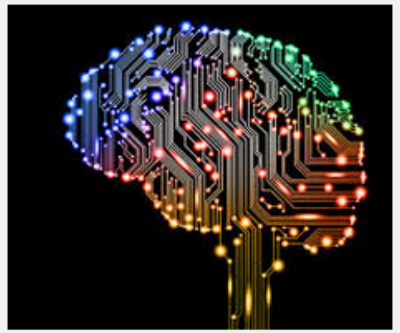
Worth taking CSC321 before CSC411? : UofT - Reddit

Google artificial intellig

Web News



Robot



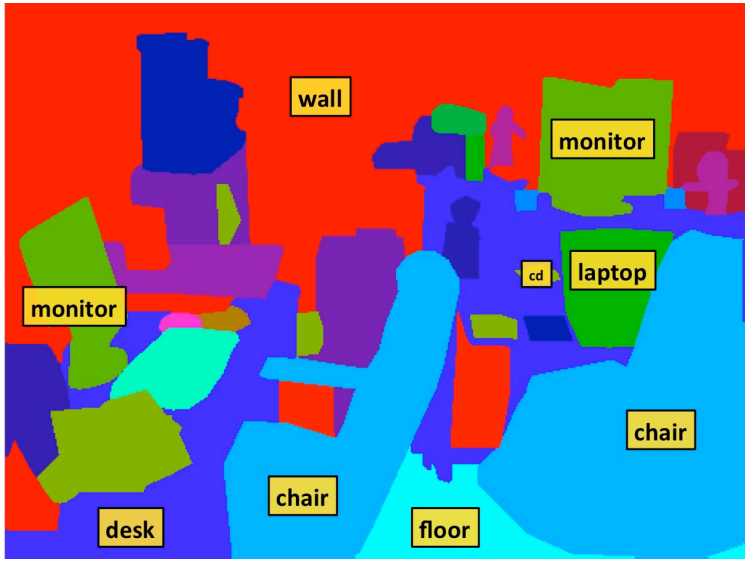


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5. Computer vision: detection, segmentation, depth estimation, optical flow, etc

Computer Vision







Learning algorithms are useful in other tasks

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6. Robotics: perception, planning, etc

Autonomous Driving







Flying Robots

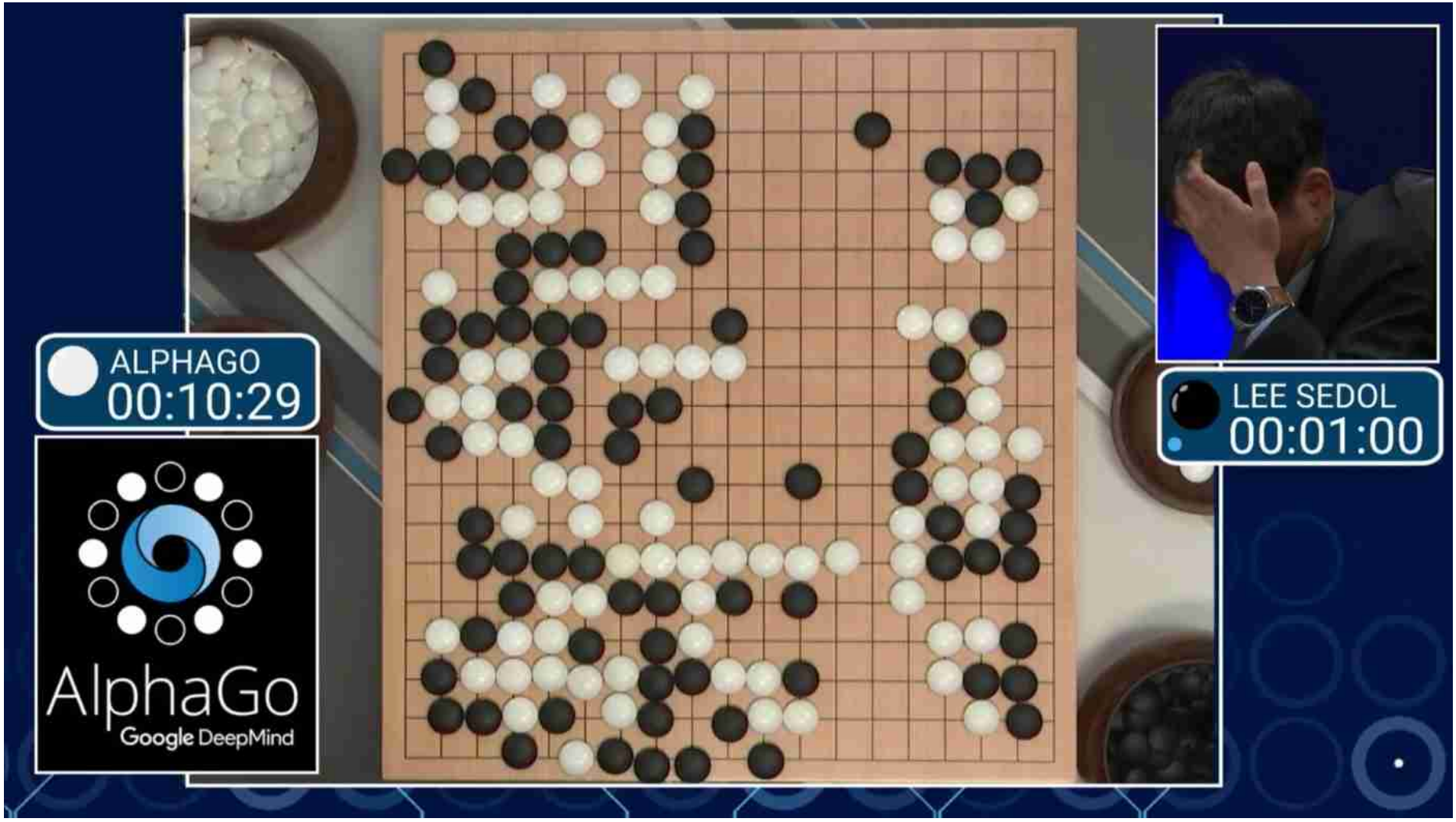


Figure : Video: https://www.youtube.com/watch?v=YQIMGV5vtd4

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7. Learning to play games

Playing Games: Alpha Go



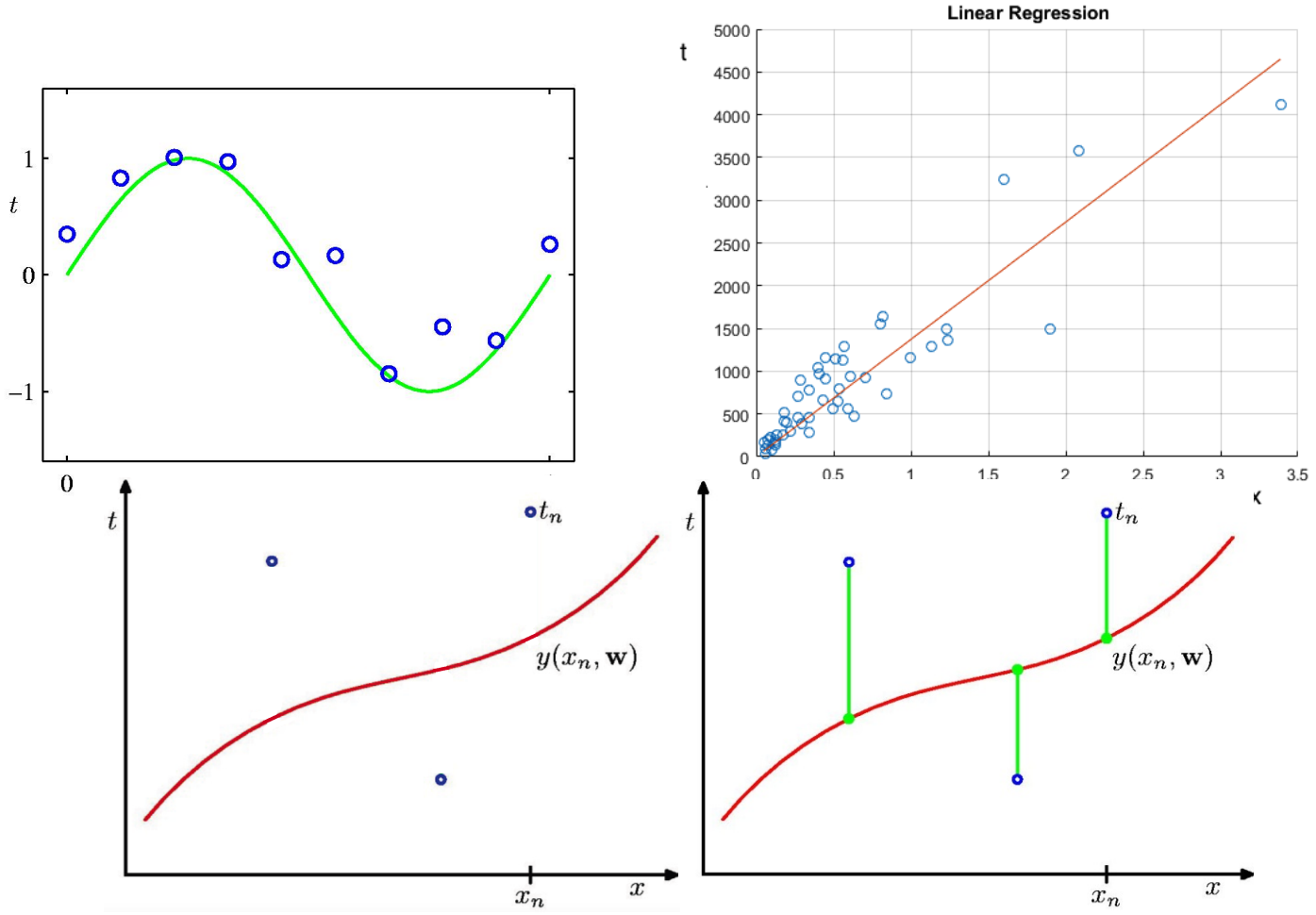
Learning algorithms are useful in other tasks

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6. Robotics: perception, planning, etc
7. Learning to play games
8. Recognizing anomalies: Unusual sequences of credit card transactions, panic situation at an airport
9. Spam filtering, fraud detection: The enemy adapts so we must adapt too
10. Many more!

Types of learning tasks

- Supervised: correct output known for each training example
  - Learn to predict output when given an input vector
    - Classification: 1-of-N output (speech recognition, object recognition, medical diagnosis)
    - Regression: real-valued output (predicting market prices, customer rating)
- Unsupervised learning
  - Create an internal representation of the input, capturing regularities/structure in data
  - Examples: form clusters; extract features
    - How do we know if a representation is good?
- Reinforcement learning
  - Learn action to maximize payoff
    - Not much information in a payoff signal
    - Payoff is often delayed

Least-Squares Regression



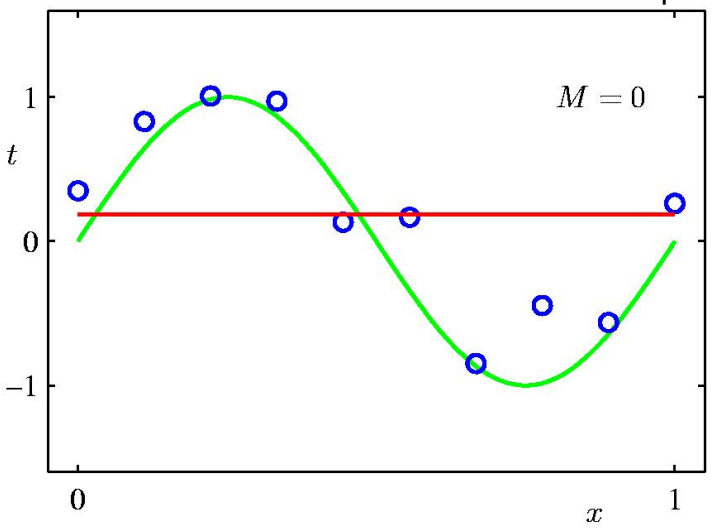
- Define a model

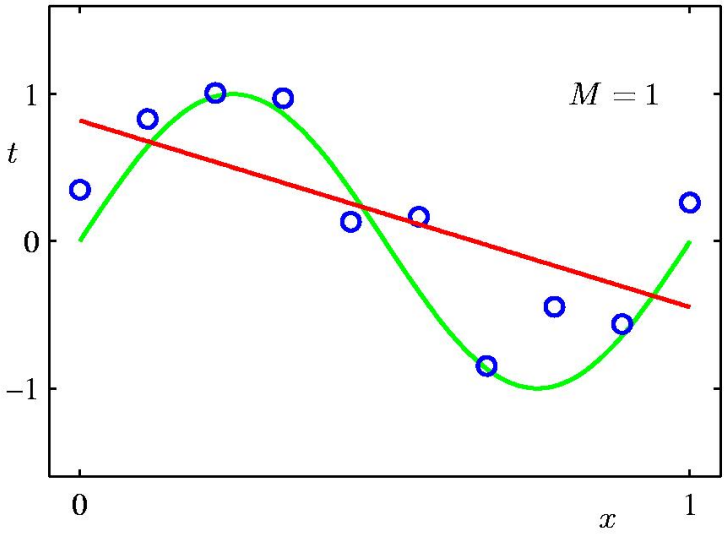
$$y(x) = \text{function}(x, \mathbf{w})$$

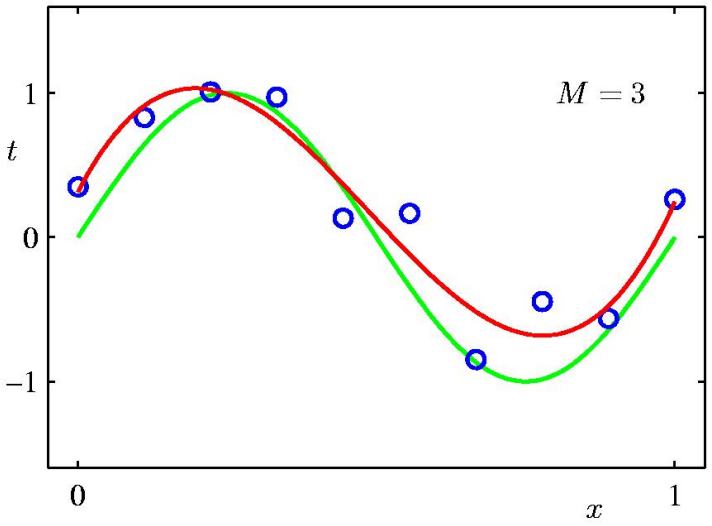
Linear: $y(x) = w_0 + w_1 x$

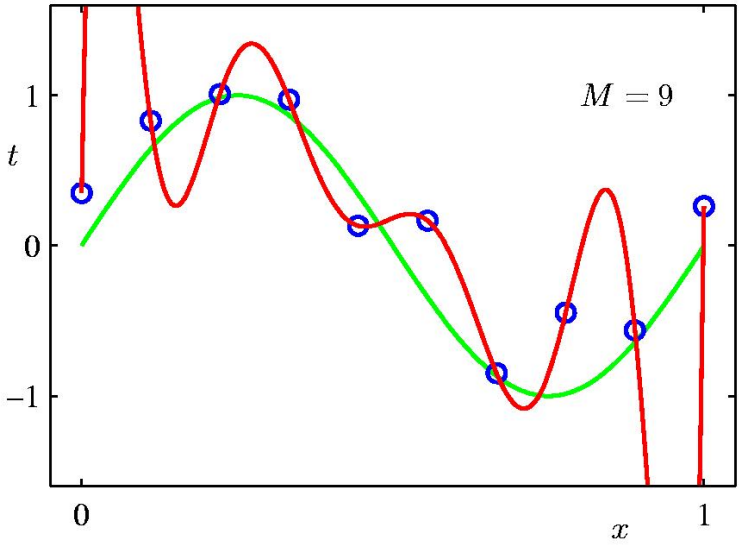
Which Fit is Best?

from Bishop









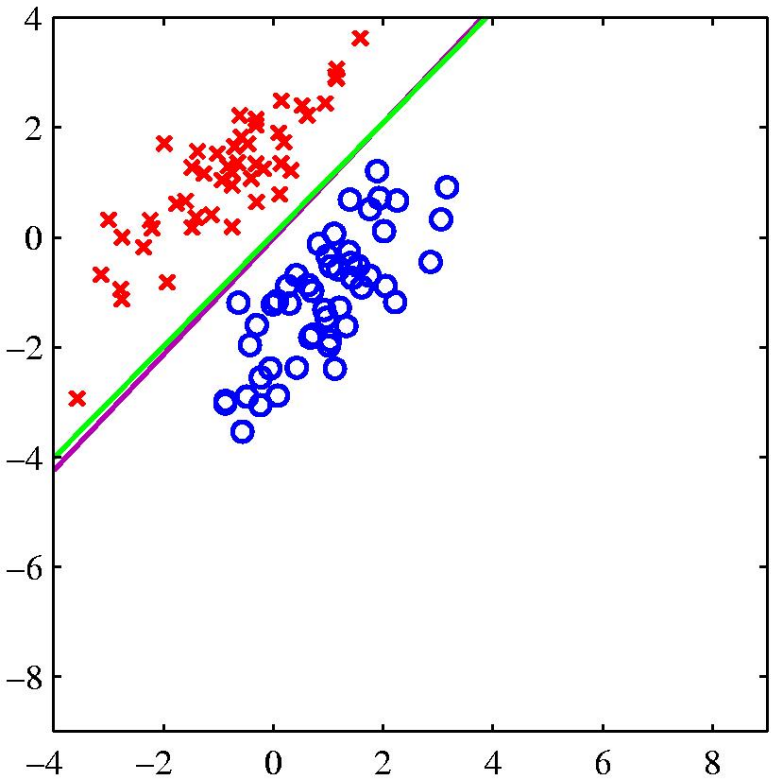
Cross-Validation (with Pictures)

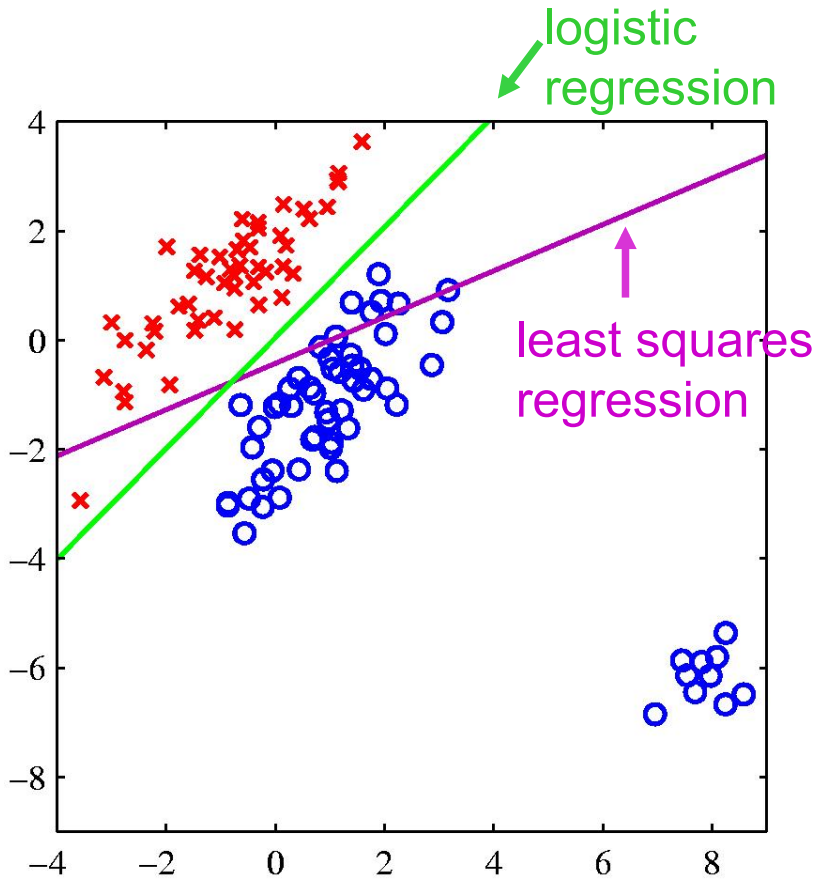
Train your model:

- Leave-one-out cross-validation:
- k-fold cross-validation:



Logistic Regression vs Least Squares Regression





If the right answer is 1 and the model says 1.5, it loses, so it changes the boundary to avoid being "too correct" (tilts away from outliers)

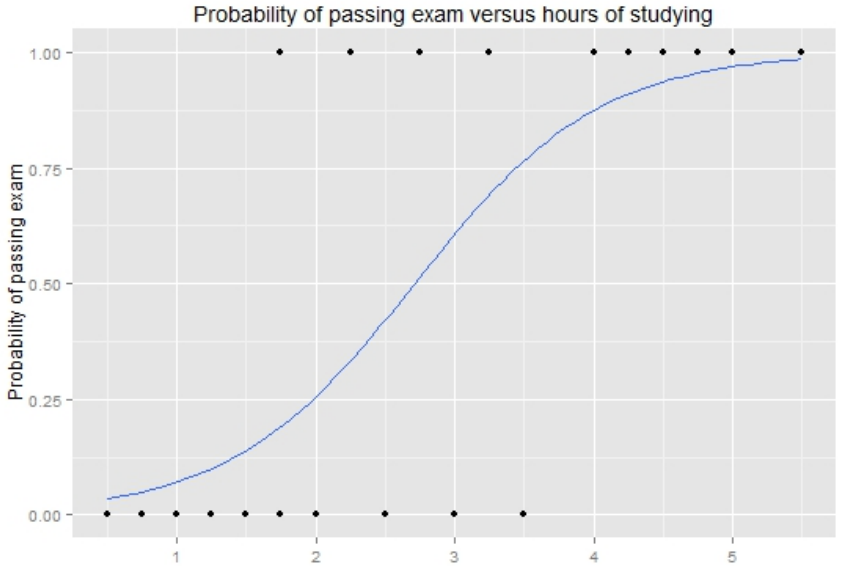
33

Example

- **Problem**: Given the number of hours a student spent learning, will (s)he pass the exam?
- Training data (top row: $x^{(i)}$, bottom row: $t^{(i)}$)

Hours	0.50	0.75	1.00	1.25	1.50	1.75	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	4.00	4.25	4.50	4.75	5.00	5.50
Pass	0	0	0	0	0	0	1	0	1	0	1	0	1	0	1	1	1	1	1	1

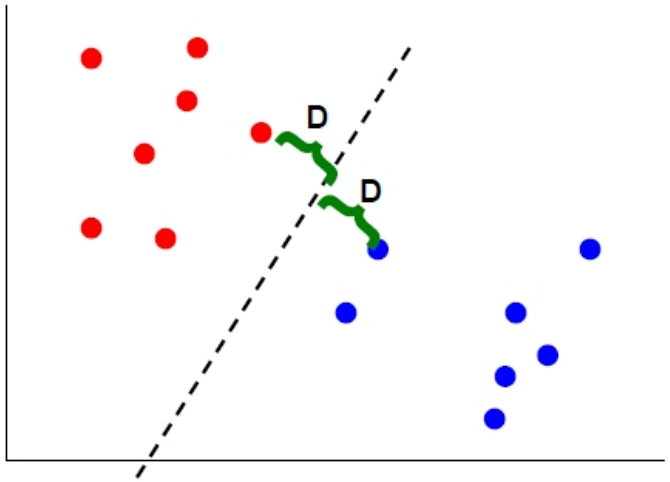
- Learn **w** for our model, i.e., logistic regression (coming up)
- Make predictions:



Hours of study	Probability of passing exam
1	0.07
2	0.26
3	0.61
4	0.87
5	0.97

Max Margin Classification

- Instead of fitting all the points, focus on the boundary points
- Aim: learn a boundary that leads to the largest margin (buffer) from points on both sides



- Why: intuition; theoretical support; and works well in practice
- Subset of vectors that support (determine boundary) are called the support vectors

Metrics

How to evaluate how good my classifier is? How is it doing on dog vs no-dog?



TP (True Positive) FP (False Positive) FN (False Negative)

PCA: Common Tool

- Handles high-dimensional data
  - If data has thousands of dimensions, can be difficult for a classifier to deal with
- Often can be described by much lower dimensional representation
- Useful for:
  - Visualization
  - Preprocessing
  - Modeling – prior for new data
  - Compression

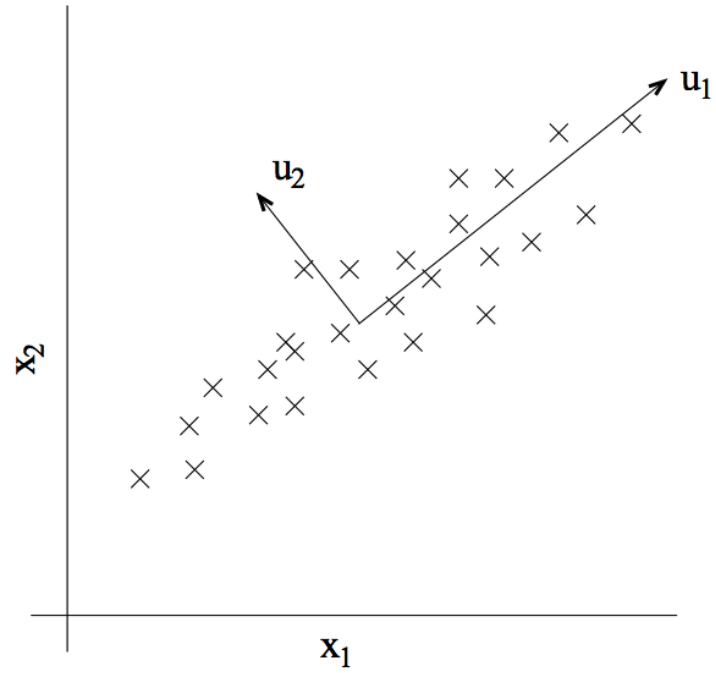
PCA: Intuition

- As in the previous lecture, training data has N vectors, $\{\mathbf{x}_n\}_{n=1}^N$, of dimensionality D, so $\mathbf{x}_i \in \mathbb{R}^D$
- Aim to reduce dimensionality:
  - linearly project to a much lower dimensional space, $M << D$:

$$\mathbf{x} \approx U\mathbf{z} + \mathbf{a}$$

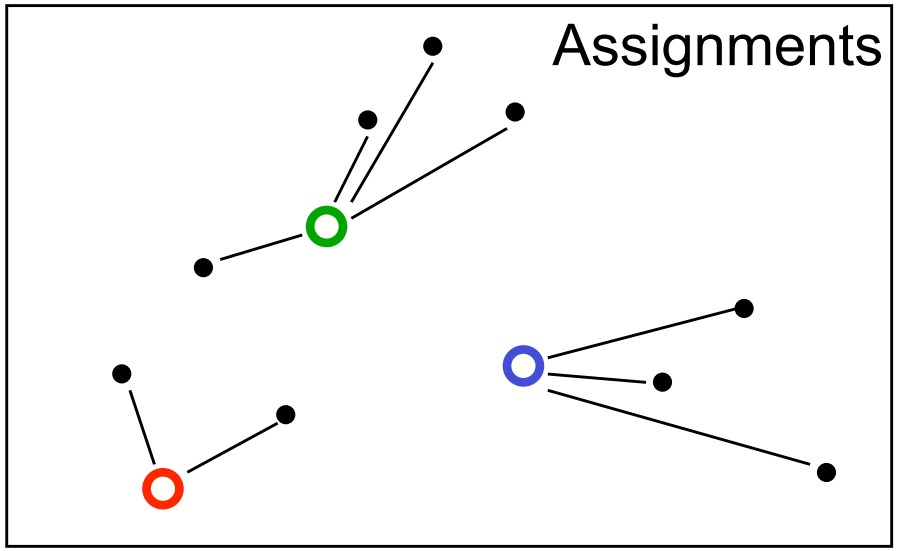
where U is a $D \times M$ matrix and $\mathbf{z}$ a M-dimensional vector

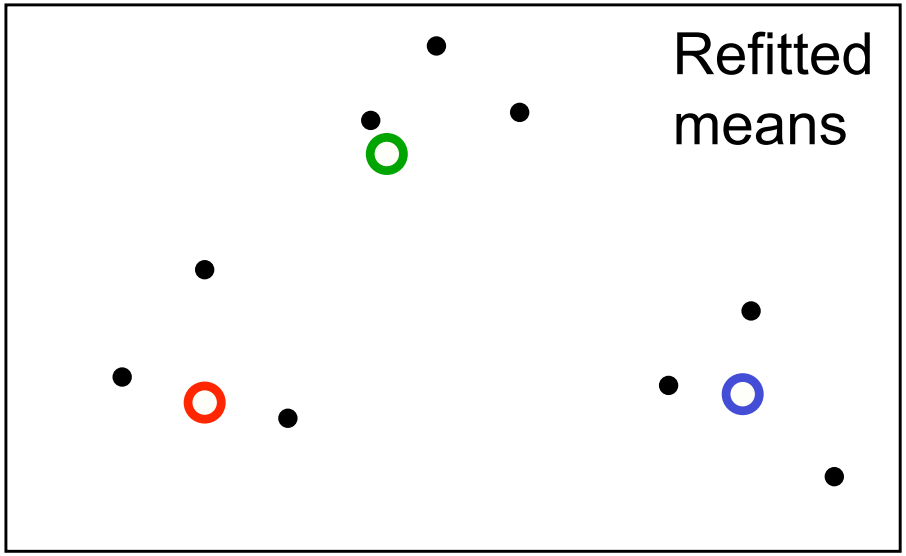
- Search for orthogonal directions in space with the highest variance
  - project data onto this subspace
- Structure of data vectors is encoded in sample covariance



K-means

- Initialization: randomly initialize cluster centers
- The algorithm iteratively alternates between two steps:
  - Assignment step: Assign each data point to the closest cluster
  - Refitting step: Move each cluster center to the center of gravity of the data assigned to it





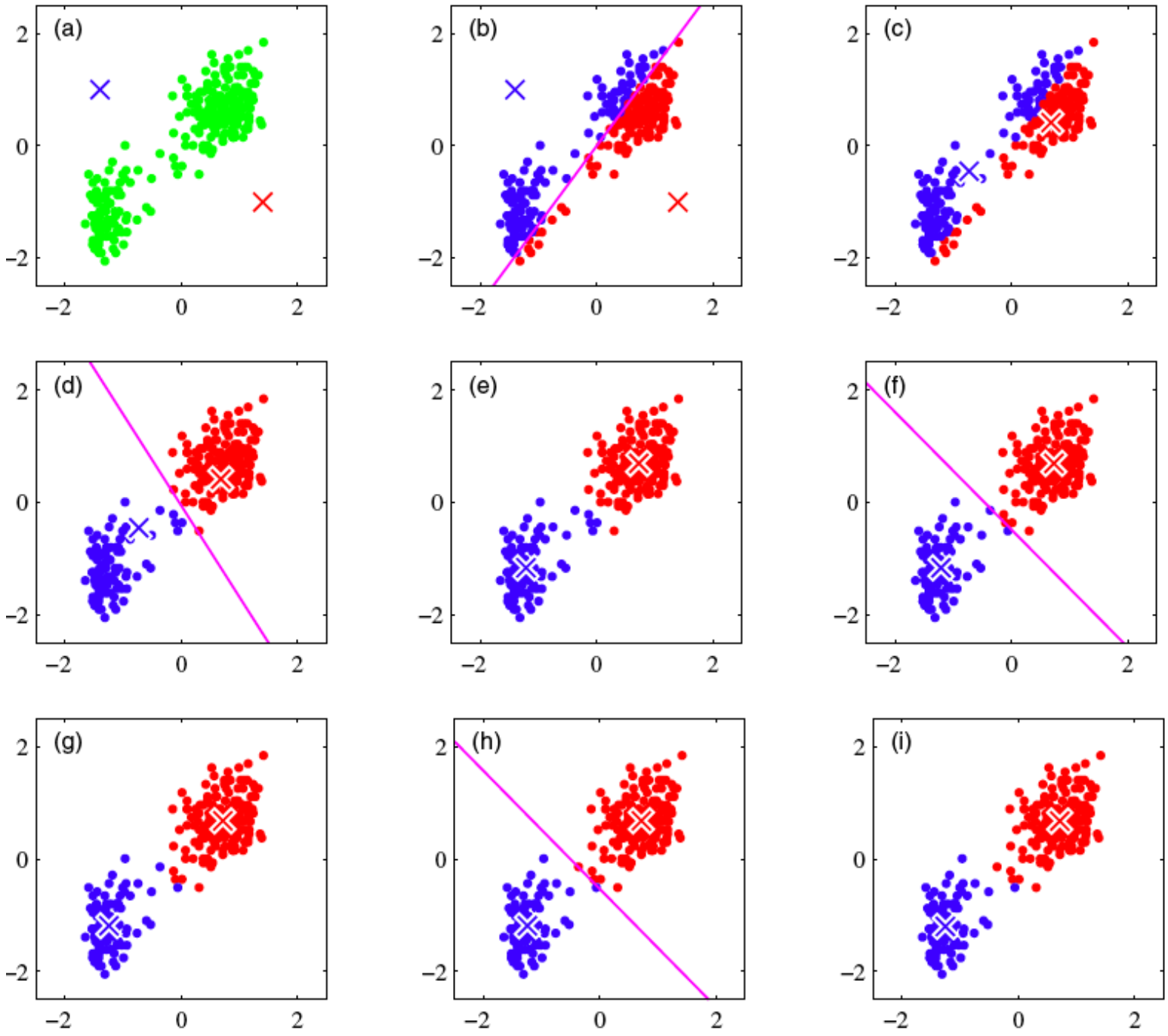


Figure from Bishop

Simple demo: http://syskall.com/kmeans.js/